



RIVER VALLEY HIGH SCHOOL

H2 CHEMISTRY

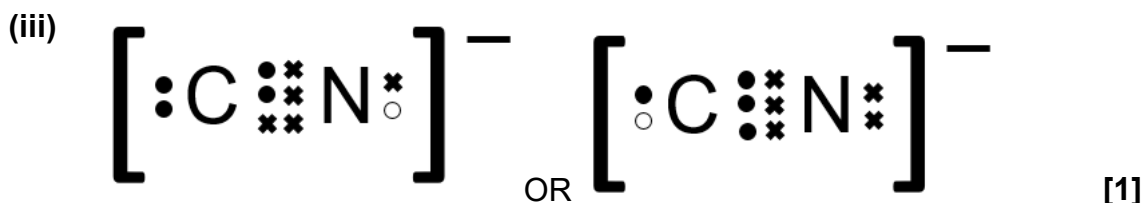
Suggested Solutions

Paper 1

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	C	16	B
2	D	17	C
3	B	18	D
4	D	19	D
5	A	20	B
6	B	21	C
7	B	22	B
8	A	23	A
9	D	24	D
10	B	25	C
11	C	26	D
12	C	27	A
13	D	28	C
14	C	29	A
15	B	30	D

Paper 2

- 1 (a) (i) $\text{HCN} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{CN}^-$ [1]
- (ii) $\text{NaCN} + \text{H}_2\text{O} \rightleftharpoons \text{Na}^+ + \text{HCN} + \text{OH}^-$
 (A common error showed the dissociation of NaCN.) [1]
- (iii) acid: H_2O , conjugate base: OH^-
 base: CN^- , conjugate acid: HCN
 (Confusion of the meaning of the term was seen in weaker responses; some answers were only given in terms of the NaCN species, others mixed the pairs together or identified the correct pairs whilst not assigning the acid-base nature correctly in both cases.) [1]
- (iv) $K_a = 10^{-4.79} = 1.6218 \times 10^{-5}$
 $= 1.62 \times 10^{-5} \text{ mol dm}^{-3}$ [1]
- (v) At 25°C , $\text{p}K_a + \text{p}K_b = 14$
 $\text{p}K_b = 14 - 4.79 = 9.21$ [2]
- (b) (i) Nucleophilic addition [1]
- (ii) HCN is a weak acid that dissociate partially in water to give a small amount of CN^- as the nucleophile, while KCN is a soluble salt which will give higher amount of CN^- as the nucleophiles.
 OR
 HCN is gaseous, toxic and will be more difficult to handle as compared to solid KCN.
 OR
 The negatively charged CN^- is a stronger nucleophile than HCN, and the acidic condition provides the H^+ to protonate the alkoxide in step 2.
 (The slightly acidic conditions will prevent excessive formation of HCN which will lower the initial concentration of CN^- (aq) and slow down the reaction)
 (Some weaker responses were vague; 'it is present in higher concentrations', or highlighted some misunderstanding of the reaction involved, ' H^+ is the catalyst', or focused only on the effect of adding acid to the position of the HCN and water equilibrium.) [2]



- (iv) Both C and N has a lone pair of electrons, but C is less electronegative than N, so C donates the lone pair of electrons more readily. C is the stronger nucleophilic site/ has greater nucleophilicity.

(Stronger responses gave explanations in terms of the greater ability of the carbon atom to form a coordinate bond to the carbonyl group or described a feature of the carbon atom which explained the behaviour. Weaker attempts focused on the difference in electronegativity between two atoms.) [1]

- (c) (i) In a CN^{-} ion, the C atom bears a lone pair of electrons which can be donated into the energetically accessible vacant orbital of the central gold(I) ion forming a co-ordinate (dative) bond. [2]
- (ii) $[\text{Au}(\text{CN})_2]^{-}$ [1]

[Total: 14]

- 2 (a) $\text{Al}_2\text{O}_3(\text{s})$ is insoluble in water due to the large magnitude of its lattice energy. The energy released during hydration of the ions through the formation of ion-dipole interaction is insufficient to compensate for the energy required to overcome the strong electrostatic forces of attraction between the Al^{3+} ions and O^{2-} ions to break down the giant ionic lattice.

(Weaker responses described only the high lattice energy of aluminium oxide or gave no reference to the energy, required or released, during the process of dissolving) [1]

- (b) (i) $\text{Al}_2\text{O}_3(\text{s}) + 2\text{NaOH}(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4](\text{aq})$ [2]
- (ii) $2\text{Al}(\text{s}) + 6\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Al}(\text{OH})_3(\text{s}) + 3\text{H}_2(\text{g})$
 $2\text{Al}(\text{s}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow \text{Al}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g})$

As the oxide layer is removed by $\text{NaOH}(\text{aq})$, the aluminium beneath the oxide layer is exposed and reacts with water to form hydrogen gas. [2]

- (c) (i) This is to form a layer of aluminium oxide (Al_2O_3) on the surface of the metal, making it resistant to corrosion. In the process of anodising, dilute sulfuric acid is used as the electrolyte. Oxygen gas

is produced at the anode which then oxidises the aluminium object placed as the electrode to produce aluminium oxide.

[2]

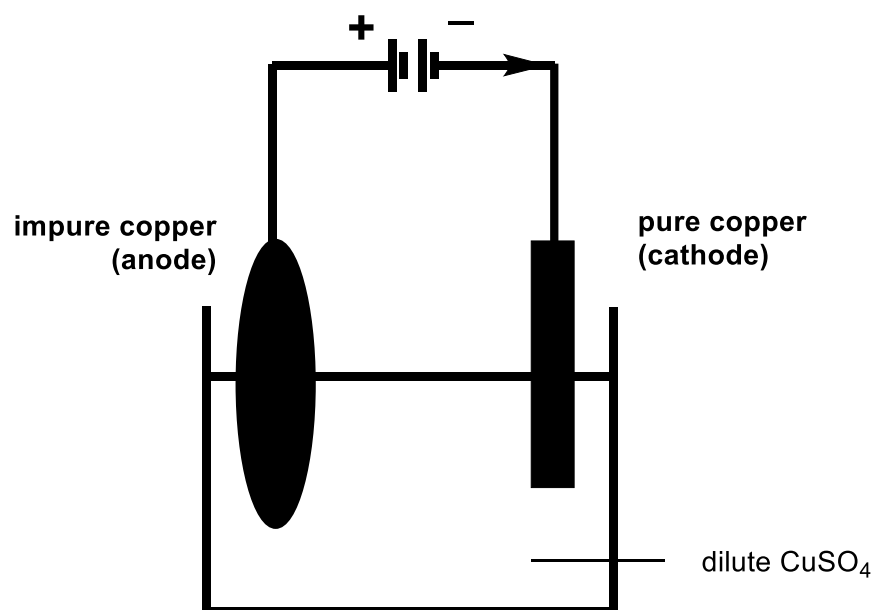
(‘Protect from rusting’ was seen relatively frequently to convey the idea that the aluminium becomes more resistant to corrosion even though ‘rusting’ is a term specific to the corrosion of iron. Some weaker answers described anodising as a process that involved coating aluminium with silver or zinc or was a technique used to purify aluminium.)

(ii)

	type of reaction occurring	half-equation(s)
anode	oxidation	$2\text{H}_2\text{O(l)} \rightarrow \text{O}_2\text{(g)} + 4\text{H}^+\text{(aq)} + 4\text{e}^-$
cathode	reduction	$2\text{H}^+\text{(aq)} + 2\text{e}^- \rightarrow \text{H}_2\text{(g)}$

[2]

(d)



[1]

(e) (i) Al^{3+} : $1\text{s}^22\text{s}^22\text{p}^6$ Cr^{3+} : $1\text{s}^22\text{s}^22\text{p}^63\text{s}^23\text{p}^63\text{d}^3$

[2]

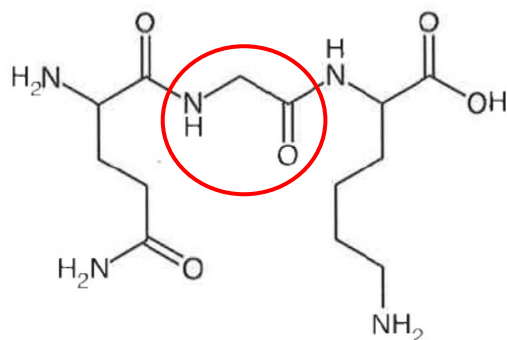
(ii) Cr^{3+} has a larger proton number than Al^{3+} (higher nuclear charge). The additional inner principal quantum shell of electrons in Cr^{3+} largely cancels out its higher nuclear charge. Therefore, the effective nuclear charge is similar, giving rise to similar ionic radii of the Cr^{3+} and Al^{3+} ions.

[2]

[Total: 14]

3 (a) Condensation [1]

(b)



W

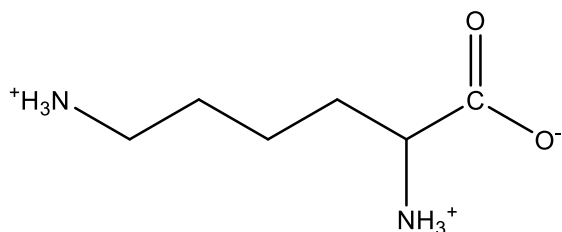
Fig. 3.1

(Some candidates selected the correct central part of **W** but included two amine groups.)

X does not rotate plane-polarised light hence **X** contains an achiral carbon with two hydrogen atoms bonded to it and has a plane of symmetry. [2]

(c) A zwitterion refers to a species that has both positively and negatively-charged groups. [1]

(d) (i)



(ii) pH 10 (any pH between 9.16 and 10.67) [1]

(e) (i) **Y**: amine

Z: amide [1]

(ii) In amines, the lone pair of electrons on its N atom is available for protonation. Therefore, it forms a conjugate acid that has a K_a value.

The p orbital of N atom overlaps with the π electron cloud of the neighbouring carbonyl group ($C=O$). The lone pair of electrons on N atom is delocalised into the $C=O$ group and hence is not available for protonation. [2]

(f)

T	U	V
$\begin{array}{c} \text{H}_2\text{N}-\text{CH}-\text{CH}_2\text{OH} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2\text{NH}_2 \end{array}$	$\begin{array}{c} \text{O} \\ \\ ^+\text{H}_3\text{N}-\text{CH}-\text{C}-\text{OH} \\ \\ \text{CH}_2 \\ \\ \text{C}=\text{O} \\ \\ \text{OH} \end{array}$	$\begin{array}{c} \text{H}_2\text{N}-\text{CH}-\text{COO}^- \\ \\ \text{CH}_2 \\ \\ \text{C}=\text{O} \\ \\ \text{NH}_2 \end{array}$

(Examiner's report: The most common correct structure shown was for **V**. Many structures of **U** highlighted the common misconception that amides do not undergo hydrolysis with acid and/or did not show the protonated amine group in acid conditions. Deducing structure **T** proved most challenging. Incorrect structures included those with incorrect carbon backbones or with two or three hydroxyl groups.

[3]

(g) Step a: acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, heat with immediate distillation

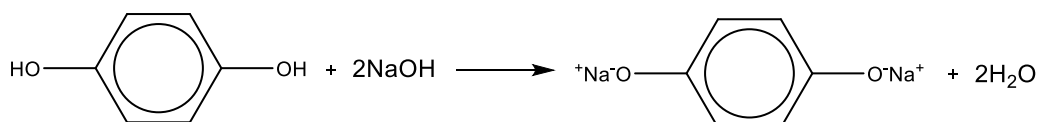
Step b: dilute aqueous sulfuric acid, heat under reflux; followed by the careful addition of an alkali.

[3]

[Total: 16]

4

(a)



[1]

(b) Amount of Na used = $\frac{0.30}{23.0} = 0.0130 \text{ mol}$

Phenol is the most acidic and reacts most vigorously/ reacts the fastest/ produces most gas per unit time to produce 0.0050 mol (a lower volume) of a colourless odourless gas [that extinguishes a lighted splint with a 'pop' sound (H_2 gas)].

Hydroquinone is less acidic and reacts vigorously to produce 0.0065 mol (a higher volume) of a colourless odourless gas [that extinguishes a lighted splint with a 'pop' sound (H_2 gas)].

Ethanol is the least acidic and reacts less vigorously/ reacts the slowest/ produces least gas per unit time to produce 0.0050 mol (a lower volume) of a colourless odourless gas [that extinguishes a lighted splint with a 'pop' sound (H_2 gas)].

OR

Measure time taken to obtain 0.0050 mol of gas. Phenol will take the least time as it is the most acidic, followed by hydroquinone, then ethanol.

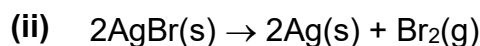
[2]

[In the suggested answer above, we proposed that hydroquinone is less acidic due to the delocalisation of electrons from O in –OH, which will reduce the delocation of negative charge on O[–] into the benzene ring. Under real conditions, hydroquinone (pK_a = 9.9) is actually slightly more acidic than phenol (pK_a = 9.95) due to the electronegative O increasing the delocalisation of negative charge on O[–] into the benzene ring. For this question, it is unlikely that the examiners were expecting candidates to know this]

- (c) (i) Silver bromide is an ionic compound and has a giant ionic lattice structure with strong electrostatic forces of attraction between oppositely-charged Ag⁺ and Br[–] ions. Therefore, AgBr has a relatively high melting point and requires sufficiently high energy to overcome the ionic bonds.

(Examiner's report: Occasionally, an answer did not refer to the structure of silver bromide or referred to intermolecular forces of attraction between silver bromide molecules.)

[2]



[1]

- (iii) Light of lower frequency has lower energy.

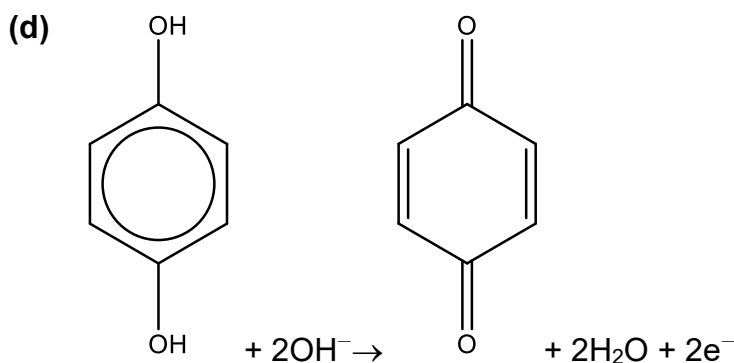
In Br[–], the valence electrons are less attracted to the Br[–] nucleus due to the additional principal quantum shell of electrons than Cl[–]. The ionic radius of Br[–] is larger and the additional inner shells of electrons provide better shielding.

Therefore, less energy will be required to remove an electron from Br[–] than Cl[–].

[3]

- (iv) Dark condition is necessary to preserve the hidden image. Overexposure to light will cause the AgBr crystal without the hidden image to undergo redox reaction and turn black.

[1]

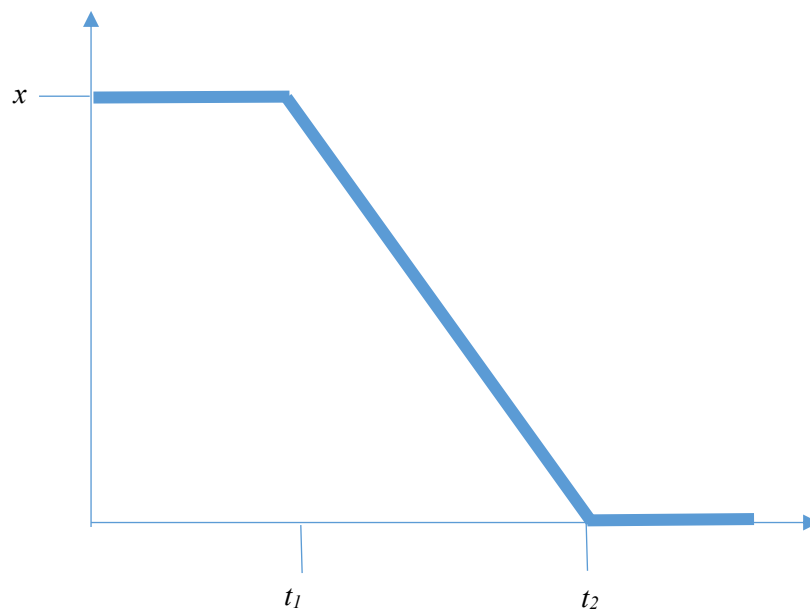


[2]

- (e) (i) Catalyst

[1]

(ii)



[2]

[Total:15]

5 (a) (i)

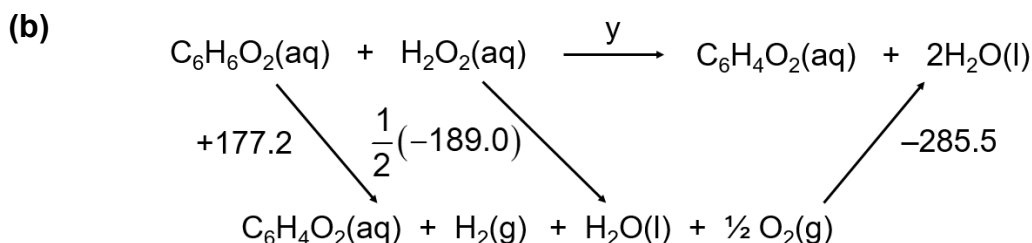
species	$\text{C}_6\text{H}_6\text{O}_2$	H_2O_2
mass ratio	10	25
mole ratio	$\frac{10}{110} = 0.0909$	$\frac{25}{34} = 0.7353$
simplest ratio	1	8

 H_2O_2 is in excess (shown).

[2]

- (ii) As more $\text{O}_2(\text{g})$ molecules are produced from the decomposition of H_2O_2 , the frequency of collisions between the molecules and the wall of the reaction chamber increases, thus the pressure inside the chamber increases.

[1]



$$y = +177.2 + \frac{1}{2}(-189) + (-285.5)$$

$$y = -202.8 = -203 \text{ kJ mol}^{-1}$$

[2]

(c) Mass of $\text{C}_6\text{H}_6\text{O}_2$ in the reaction mixture = $0.1 \times 1.0 \times 10^{-3} = 1.0 \times 10^{-4} \text{ g}$

$$n_{\text{C}_6\text{H}_6\text{O}_2} = \frac{1.0 \times 10^{-4}}{6(12) + 6(1) + 2(16)} = 9.0909 \times 10^{-7} \text{ mol}$$

Energy released from reaction in equation 1

$$= 9.0909 \times 10^{-7} \times 202.8 = 1.8436 \times 10^{-4} \text{ kJ}$$

Amount of excess H_2O_2

$$= \left(\frac{0.25 \times 1 \times 10^{-3}}{2(16) + 2(1)} \right) - 9.0909 \times 10^{-7} = 6.4439 \times 10^{-6} \text{ mol}$$

Energy released from decomposition of excess H_2O_2

$$= 6.4439 \times 10^{-6} \times \frac{189}{2} = 6.0849 \times 10^{-4} \text{ kJ}$$

Total amount of energy released

$$= 1.8436 \times 10^{-4} + 6.0849 \times 10^{-4} = 7.9285 \times 10^{-4} = 7.93 \times 10^{-4} \text{ kJ}$$

[4]

(d) (i) $\Delta G_{\text{eqn1}} = -206.5 \text{ kJ mol}^{-1}$. Since $\Delta G_{\text{eqn1}} < 0$, the reaction is spontaneous.

[1]

(ii) In the reservoir, the reaction has a large energy barrier/ high activation energy. The reaction chamber contains the enzymes catalase and peroxidase which provided an alternative pathway with lower activation energy for the reaction in equation 1 to occur quickly. More molecules have energy that is greater or equal to the activation energy, which leads to higher frequency of effective collision, faster rate of reaction.

(Weaker answers indicated that the information given in the text was not fully understood regarding the location and effects of the enzymes.)

[2]

(e) At high $[\text{H}_2\text{O}_2]$, all the active sites on the enzyme molecules are saturated/ occupied. The rate of reaction then becomes independent of $[\text{H}_2\text{O}_2]$ and the reaction is zero-order with respect to the H_2O_2 .

[1]

Some responses were vague, for example 'concentration of hydrogen peroxide is not rate limiting' or 'there is not enough catalyst'. A minority seemed not to understand the process and answered in terms of 'all the reactant is used up'.

(f) $E^\ominus(\text{H}_2\text{O}_2/\text{H}_2\text{O}) = +1.77\text{V}$

$$\Delta G = -nFE$$

$$E^\ominus_{\text{cell}} = \frac{-206.5 \times 1000}{-(2)(96500)} = +1.0699\text{V}$$

$$E^\ominus_{\text{cell}} = E^\ominus(\text{H}_2\text{O}_2/\text{H}_2\text{O}) - E^\ominus(\text{C}_6\text{H}_4\text{O}_2/\text{C}_6\text{H}_6\text{O}_2)$$

$$E^\ominus(\text{C}_6\text{H}_4\text{O}_2/\text{C}_6\text{H}_6\text{O}_2) = +1.77 - (+1.6099) = +0.700\text{V}$$

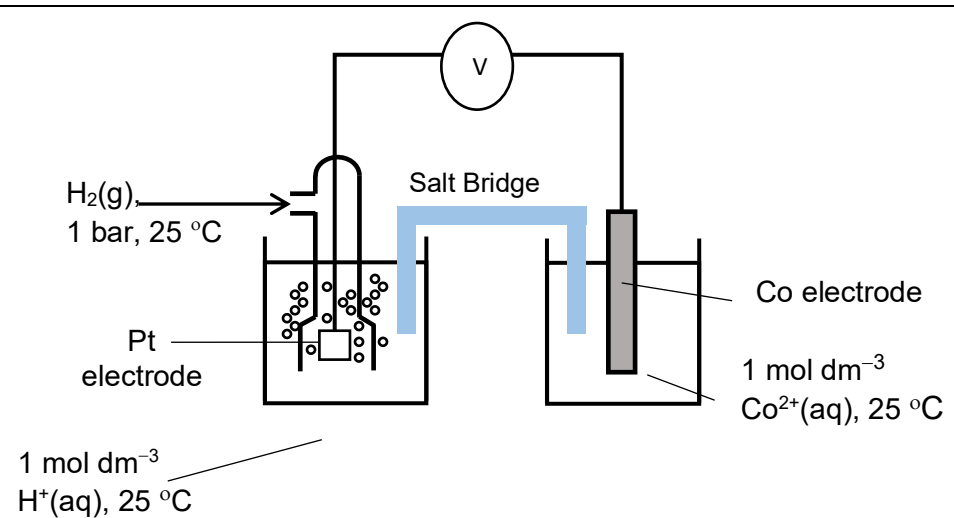
(Errors in this calculation included use of an incorrect formula, using a value of $n = 1$ or $n =$ amount of hydroquinone in the reaction mixture or not converting the value of ΔG to J mol^{-1} .)

[3]

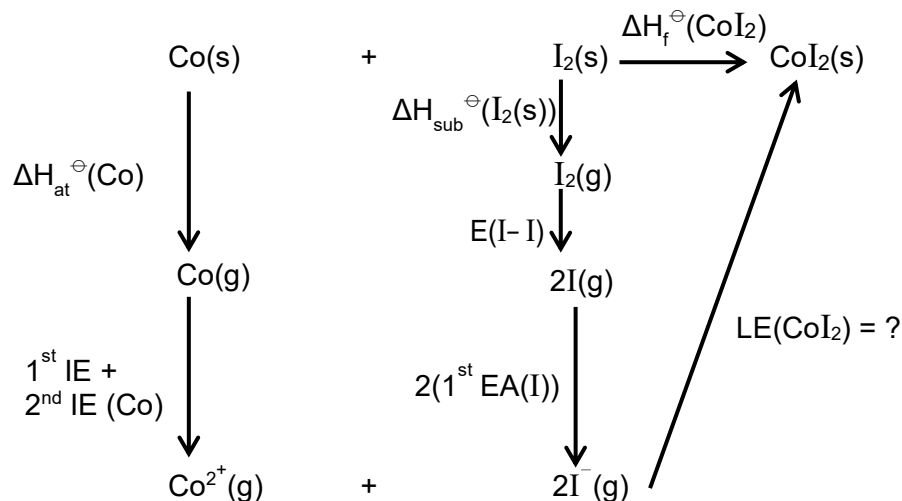
[Total: 16]

Paper 3 Section A

1	(a)	(i)	A transition element is a d-block element which forms at least one stable ion with a partially filled d subshell.	[1]
		(ii)	Let the percentage abundance of ^{63}Cu be $x\%$ and the percentage abundance of ^{65}Cu be $(100-x)\%$. $63.546 = (62.930 \times \frac{x}{100}) + (64.928 \times \frac{100-x}{100})$ $x = 69.17\%$ $\therefore ^{63}\text{Cu} = 69.2\%, ^{65}\text{Cu} = 30.8\%$ Examiner's report : The most common error was to use 63 and 65 as the isotopic masses.	[1]
	(b)	(i)	In a complex ion, the <u>presence of ligands</u> causes the <u>3d orbitals</u> to <u>split</u> into <u>two sets</u> of non-degenerate orbitals with <u>different energies</u> . The <u>difference in energies</u> , ΔE , between the two sets of 3d orbitals is relatively <u>small</u> and falls within the energy range of the <u>visible spectrum</u> of the electromagnetic spectrum. Thus, radiation in the visible region is absorbed when <u>an electron from a lower energy d orbital is promoted to a partially filled/ empty d orbital of higher energy</u> . d-d transitions occurred. The colour observed is the <u>complement</u> of the colours absorbed.	[3]

	(ii)	Electronic configuration of Cu^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$ Electronic configuration of Cu^+ : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$ Cu^{2+} has a <u>partially filled 3d subshell where d-d transitions (electronic transitions) can occur</u>. Cu^{2+} is coloured. However, for Cu^+ has a <u>completely filled 3d subshell and d-d transitions (electronic transitions) cannot occur</u>. Hence Cu^+ is colourless.	[1]
(c)	(i)	 Examiner's report : Common errors were the omission of the salt bridge and a temperature (298 K).	[3]
	(ii)	From the data booklet : $\text{Co}^{2+} + 2e \rightleftharpoons \text{Co} \quad E^\ominus = -0.28\text{V}$ $\text{Fe}^{3+} + e \rightleftharpoons \text{Fe}^{2+} \quad E^\ominus = +0.77\text{V}$ When the mixture of Fe^{2+} and Fe^{3+} is added to the cobalt half cell, Fe^{3+} will be reduced to Fe^{2+} while Co will be oxidised to Co^{2+}. $\text{Co} + 2\text{Fe}^{3+} \rightarrow 2\text{Fe}^{2+} + \text{Co}^{2+}$ $E^\ominus_{\text{cell}} = 0.77 - (-0.28) = 1.05\text{V}$	[2]
(d)	(i)	Lattice energy is the enthalpy change when one mole of the solid ionic compound is formed from its constituent gaseous ions under standard conditions. Examiner's report : Common errors were stating "energy required"; omitting 1 mole with the ionic compound formed and starting from 1 mole of gaseous ions.	[1]

(ii)



$$\Delta H_{\text{at}}^{\ominus}(\text{Co}) + 1^{\text{st}} \text{ IE} + 2^{\text{nd}} \text{ IE (Co)} + \Delta H_{\text{sub}}^{\ominus}(\text{I}_2(\text{s})) + \text{BE (I-I)} + 2(1^{\text{st}} \text{ EA(I)}) + \text{LE(CoI}_2) = \Delta H_{\text{f}}^{\ominus}(\text{CoI}_2)$$

$$424.7 + 757 + 1640 + 62.4 + 151 + 2(-295.4) + \text{LE} = -88.7$$

$$\text{LE} = -2533 \text{ kJ mol}^{-1}$$

$$\approx -2530 \text{ kJ mol}^{-1} \text{ (3 s.f.)}$$

- Energy level diagram can be drawn instead of Born-Haber cycle.

Examiner's report : The data extraction was good, except for the I-I bond energy of 151. Another common error involved an incorrect multiplier with the electron affinity of iodine.

[3]

(iii) The order of lattice energy in decreasing magnitude is $\text{CoO} > \text{CoF}_2 > \text{CoI}_2$

$$|\text{Lattice energy}| \propto \left| \frac{q^+ q^-}{r^+ + r^-} \right|$$

The charge and size of the cation, Co^{2+} , is the same for all 3 compounds. O^{2-} has the greatest charge. F^- is the smallest, followed by O^{2-} and I^- is the largest. The multiplication of the charges has a greater effect than the sum of interionic radius. Hence, CoO has a greater magnitude of lattice energy than CoF_2 . CoI_2 has the smallest magnitude of lattice energy.

Examiner's report : Many candidates correctly stated the expression describing the relationship of lattice energy to the cationic (q^+) and anionic charges (q^-), and the interionic distance ($r^+ + r^-$).

Most were able to apply this knowledge to state and clearly explain their order of lattice energy. Some candidates explained in terms of

[3]

F_2^{2-} as the anion and stated CoF_2 had the most exothermic lattice energy.

(e) Formula of compound **A** : $\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3$

Formula of compound **B** : $[\text{Co}(\text{NH}_3)_6]^{3+}[\text{Co}(\text{NO}_2)_6]^{3-}$

M_r of compound **B** = $247.9 \times 2 = 495.8$

Molecular formula of **B** = $\text{Co}_2\text{H}_{18}\text{N}_{12}\text{O}_{12}$

1) Oxidation number of Co in **A**

Let the oxidation number of Co be x

$$x + 0 + 3(-1) = 0$$

$$x = +3$$

2) Oxidation number of Co in cation of compound **B**

Let the oxidation number of Co be x

$$x + 0 = +3$$

$$x = +3$$

3) Oxidation number of Co in anion of compound **B**

Let the oxidation number of Co be x

$$x + 6(-1) = -3$$

$$x = +3$$

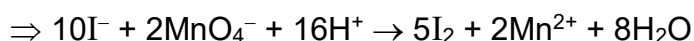
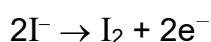
[3]

(f) (i) $2\text{MnO}_4^- + 6\text{H}^+ + 5\text{NO}_2^- \rightarrow 2\text{Mn}^{2+} + 5\text{NO}_3^- + 3\text{H}_2\text{O}$

[1]

(ii) $\text{Amt of } \text{S}_2\text{O}_3^{2-} = \frac{22.40}{1000} \times 0.0150 = 3.360 \times 10^{-4} \text{ mol}$

$$\text{Amt of } \text{I}_2 = 3.360 \times 10^{-4} \div 2 = 1.680 \times 10^{-4} \text{ mol}$$



$$\text{Amt of excess } \text{MnO}_4^- = (1.680 \times 10^{-4} \div 5) \times 2 = 6.720 \times 10^{-5} \text{ mol}$$

Amt of MnO_4^- that reacted with NO_2^-

$$= \left(\frac{50.00}{1000} \times 0.00500 \right) - 6.720 \times 10^{-5} = 1.828 \times 10^{-4} \text{ mol}$$

$$\text{Amt of } \text{NO}_2^- \text{ in } 1.00\text{g sample} = (1.828 \times 10^{-4} \div 2) \times 5$$

$$= 4.570 \times 10^{-4} \text{ mol}$$

[3]

$$\begin{aligned}\text{Mass of NaNO}_2 \text{ in 1.00g sample} &= 4.570 \times 10^{-4} \times (23.1 + 14.0 + 32.0) \\ &= 0.03158\text{g}\end{aligned}$$

$$\% \text{ by mass of NaNO}_2 = \frac{0.03158}{1.00} \times 100\% = 3.16\% \text{ (Ans)}$$

Examiner's report : The most common errors were the omission of the initial amount of MnO_4^- and calculating the amount of MnO_4^- reacting with NO_2^- . Many were able to multiply their amount of MnO_4^- (reacted in step 1) by the reacting ratio and by the correct Mr to give the percentage of NaNO_2 .

[Total: 25]

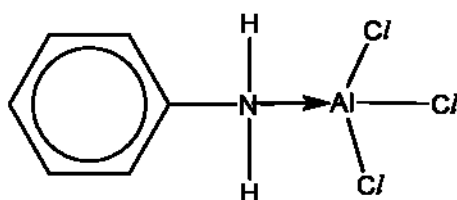
2 (a)

ethanamide	<	phenylamine	<	ethylamine
least basic				most basic
Lone pair of electrons on N is delocalised into the C=O of amide functional group. It is least available for donation.		Lone pair of electrons is delocalised into the benzene ring. It is less available for donation.		$-\text{C}_2\text{H}_5$ is electron-donating, making the lone pair of electrons on N more available for donation.

[3]

It was well known that ethylamine is the stronger base and some good explanations were seen. Common errors were stating that phenylamine was less basic than ethanamide and omitting what the lone pair on nitrogen was being made available for.

(b)



[1]

Only a minority of candidates answered this correctly.

(c) (i)

D	E

[1]

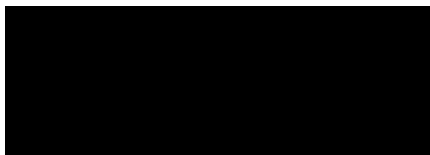
Most candidates answered this correctly. Common errors were chlorobenzene and phenylamine for **D**. Some candidates suggested a secondary amide as **E**.

- (ii) step 1: concentrated H_2SO_4 , concentrated HNO_3 , maintained at 30°C
 step 2: Sn , concentrated HCl , heat followed by NaOH (aq)
 step 3: limited ethanolic $\text{CH}_3\text{CH}_2\text{Cl}$, heat

[3]

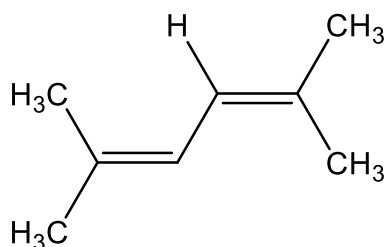
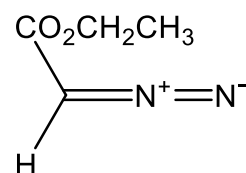
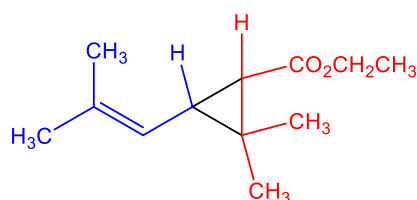
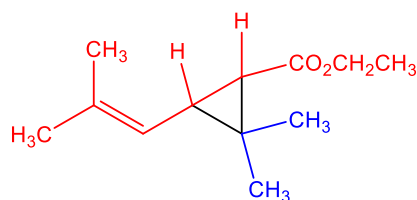
This was well known. Common errors included omitting 'concentrated' with the reagents in step 1 and stating the reagents for step 1 as CH_3Br or $\text{CH}_3\text{CH}_2\text{Br}$ with Al/Br_3 .

(d)

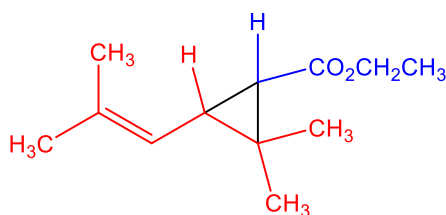


This question was generally answered well.

2 (e) (i)

Compound **G**Compound **H**

Incorrect choices for deducing **G** (in red) and **H** (in blue), as you will obtain an ester functional group when forming back **G** [Note: **G** must be a hydrocarbon].

Correct choice to deduce **G** and **H**

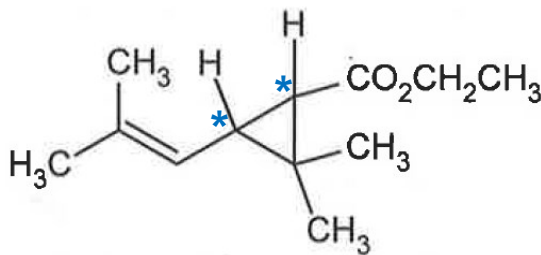
[2]

- (ii) Number of stereoisomers = **4** (i.e. 2^2)

There are **2 chiral carbons/centres** present in compound J.

[1]

(You may draw the structure of compound J out and indicate the chiral centres as your explanation.)



(Examiner's report: A common error was 8 stereoisomers.)

- (f) (i) The C—C bond energy in cyclopropane is **lower** than that in propane, due to the presence of **ring/angle strain** at all three carbon atoms in the ring. [1]

- (ii) Given that: cyclopropane → propene $\Delta H = -31 \text{ kJ mol}^{-1}$

Using the formula:

$$\Delta H_r = \Sigma E(\text{bonds broken in cyclopropane}) - \Sigma E(\text{bonds formed in propene})$$

$$-31 = [6E(\text{C—H}) + 3(289)] - [6(410) + 350 + 610]$$

$$6E(\text{C—H}) = 2522$$

$$\text{Average } E(\text{C—H}) = \underline{420 \text{ kJ mol}^{-1}} \text{ (3 s.f.)}$$

[1]

- (iii)
- With the results from each experiment, **construct a graph plot of [cyclopropane] vs time**. Then **draw a tangent at the initial point** (when time is zero) and **calculate the gradient of the tangent** to get the initial rate of that experiment.
 - Plot initial rates of each experiment against [cyclopropane]**. If the reaction is first order with respect to [cyclopropane], **a straight line that passes through origin** will be obtained.

OR

Compare the various gradients / initial rates of the experiment series with the initial [cyclopropane] used. If **the change in initial [cyclopropane] is directly proportional to the change in gradients / initial rates**, the reaction is first order with respect to [cyclopropane] (e.g. initial rate is doubled when concentration of cyclopropane is doubled).

Note: This is actually the continuous method (i.e. sampling method), with aliquots taken at timed intervals in a particular experiment. The

[2]

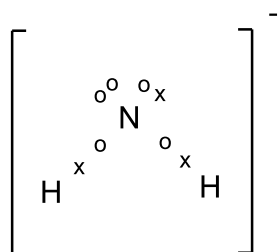
same experiment is repeated several times with different starting concentrations of cyclopropane.

- (iv) A catalyst will provide an **alternate reaction pathway of lower activation energy**, E_a' ($< E_a$), of the reaction in equation 1, hence that **increases the magnitude of k** , the rate constant. [1]

[Total: 20]

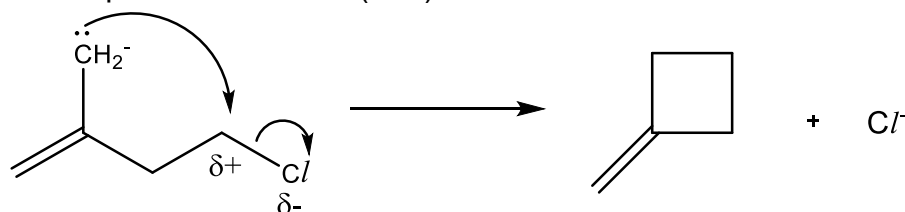
3 (a) (i) **$[\text{Na}(\text{NH}_3)_6]^+$**

The complex ion, $[\text{Na}(\text{NH}_3)_6]^+$, contains 6 NH_3 ligands which forms 6 dative covalent bonds. Around the Na^+ ion, there are 6 bond pairs of electrons and 0 lone pairs of electrons. To minimise repulsion between electron pairs, the shape is octahedral and the bond angle is 90° .

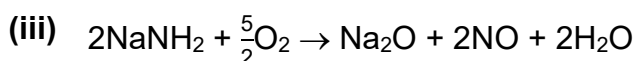


In NH_2^- , there are 2 bonds pairs and 2 lone pairs on the N atom. To minimise repulsion between electron pairs, the shape is bent and the bond angle is 104.5° . [3]

(ii) Nucleophilic substitution ($\text{S}_{\text{N}}2$)



[1]



(Examiner's report: incorrect formula of NaO was commonly used) [1]

(b) Acid strength:

butanoic acid $<$ 2-iodobutanoic acid $<$ 2-chlorobutanoic acid

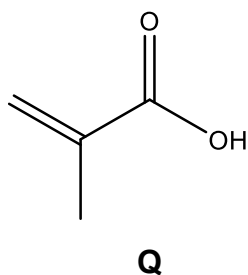
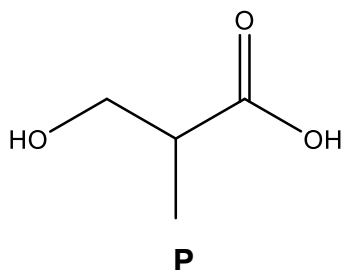
Anion/conjugate base stability:

$\text{CH}_3\text{CH}_2\text{CH}_2\text{COO}^- < \text{CH}_3\text{CH}_2\text{CH}(\text{COO}^-) < \text{CH}_3\text{CH}_2\text{CH}(\text{COO}^-)$ [2]

2-iodobutanoic acid and 2-chlorobutanoic acid are stronger acids than butanoic acid as the presence of electron-withdrawing Cl or I atoms disperses the negative charge on the COO^- group, making the substituted carboxylate anions more stable than $\text{CH}_3\text{CH}_2\text{CH}_2\text{COO}^-$.

2-chlorobutanoic acid is a stronger acid than 2-iodobutanoic acid as Cl atom, being more electronegative and hence more electron-withdrawing, disperses the negative charge on the COO^- group more effectively than I atom. Hence, $\text{CH}_3\text{CH}_2\text{CHClCOO}^-$ is a more stable anion than $\text{CH}_3\text{CH}_2\text{CHI}\text{COO}^-$.

- (c) step 1: NaOH(aq) heat under reflux
 step 2: excess concentrated H_2SO_4 , heat
 step 3: HBr(g)



[5]

- (d) Since pH is 3.28, $[\text{H}^+] = 10^{-3.28} = 5.248 \times 10^{-4} \text{ mol dm}^{-3}$
 Let $[\text{HA}]$ be x

	HA(aq)	$\rightleftharpoons$	$\text{A}^-(\text{aq})$	+	$\text{H}^+(\text{aq})$
Initial concentration/ mol dm^{-3}	x		0		0
Change in concentration/ mol dm^{-3}	-5.248×10^{-4}		$+5.248 \times 10^{-4}$		$+5.248 \times 10^{-4}$
Equilibrium concentration/ mol dm^{-3}	$x - 5.248 \times 10^{-4}$		5.248×10^{-4}		5.248×10^{-4}

Since HA is a weak acid, $x - 5.248 \times 10^{-4} \approx x$

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} = 1.38 \times 10^{-5}$$

$$\frac{(5.248 \times 10^{-4})(5.248 \times 10^{-4})}{x} = 1.38 \times 10^{-5}$$

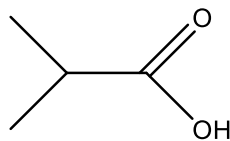
$$x = 0.01996 \text{ mol dm}^{-3}$$

Let m be the M_r of HA

[3]

$$\frac{1.76}{\frac{m}{1}} = 0.01996$$

$$m = 88.2$$



Suggested structure of HA:

(Examiner's report: Fewer candidates gave the correct structure of the branched-chain carboxylic acid. Butanoic acid was a common error)

[Total:15]

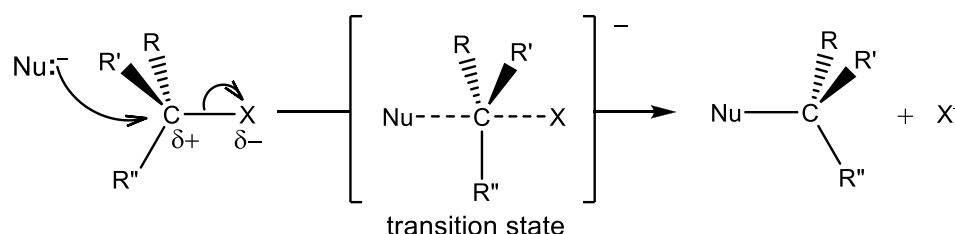
- 4 (a) (i) 1) S_N2 .
2) S_N1 .

The S_N2 mechanism is favoured by 1-bromobutane, a primary alkyl halide while the S_N1 mechanism is favoured by 2-bromo-2-methylpropane, a tertiary alkyl halide. This is due to the stability of the carbocation intermediate. In the S_N1 mechanism, the carbocation intermediate is formed after the first step. Since tertiary carbocations are more stable than primary carbocations, the tertiary alkyl halides will favour the S_N1 reaction mechanism. In the S_N2 mechanism, the 'back-side' attack of the alkyl halide occurs and is not favoured by tertiary alkyl halides due to steric hindrance.

[Identification of the type of reaction occurring when the halogenoalkane reacts with aqueous sodium hydroxide was well known. Most candidates correctly linked the relative steric hindrance of the haloalkane to the mechanism. Fewer were able to provide an explanation in terms of the stability of carbocation.]

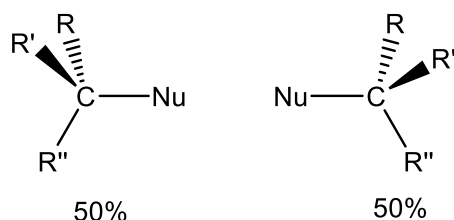
[3]

- (ii) For the S_N2 mechanism, the inversion of configuration due to the 'back-side' attack of the nucleophile to obtain a single enantiomer.



For the S_N1 mechanism, there is equal probability of attack by the nucleophile from either face of the trigonal plane around the positively charged carbon. This gives rise to equal proportions of both enantiomers. A racemic mixture is obtained.

[2]



This was generally well answered. Some confused the S_N1 and S_N2 mechanisms, stating a single enantiomer is formed via the S_N1 mechanism and a racemic mixture via the S_N2 mechanism.

(b) $\Delta H^\ominus_{\text{solution}} = 2(-230) + (+64.8) - (-449.8) = +54.6 \text{ kJ mol}^{-1}$

$$\Delta G^\ominus_{\text{solution}} = 54.6 - 298 \left(\frac{-196.2}{1000} \right) = +113 \text{ kJ mol}^{-1} (>0, \text{non-spontaneous})$$

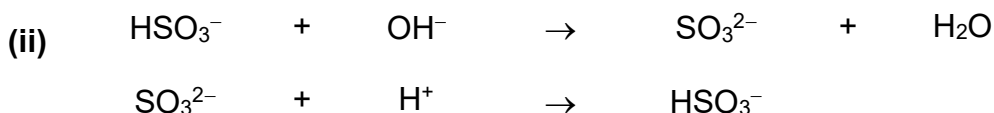
This calculation was competently done. The most common error was the omission of $\times 2$ multiplier for $\Delta H^\ominus_{\text{sol}}$. Most candidates could correctly link their $\Delta G^\ominus_{\text{sol}}$ value to solubility of $\text{Cu}(\text{OH})_2(\text{s})$.

[3]

- (c) (i) It is a solution that resists changes in pH when small amount of acid or base is added to it.

(This definition was well known by many; some did not appreciate that a buffer solution resists changes in pH only if small amounts of acid or alkali are added.)

[1]



(Many candidates were able to write two equations to illustrate how this solution acts as a buffer.)

[1]

(iii) Amount of HSO_3^- left = $\frac{50.0}{1000} \times 0.500 - \frac{25.0}{1000} \times 0.600 = 0.1000 \text{ mol}$

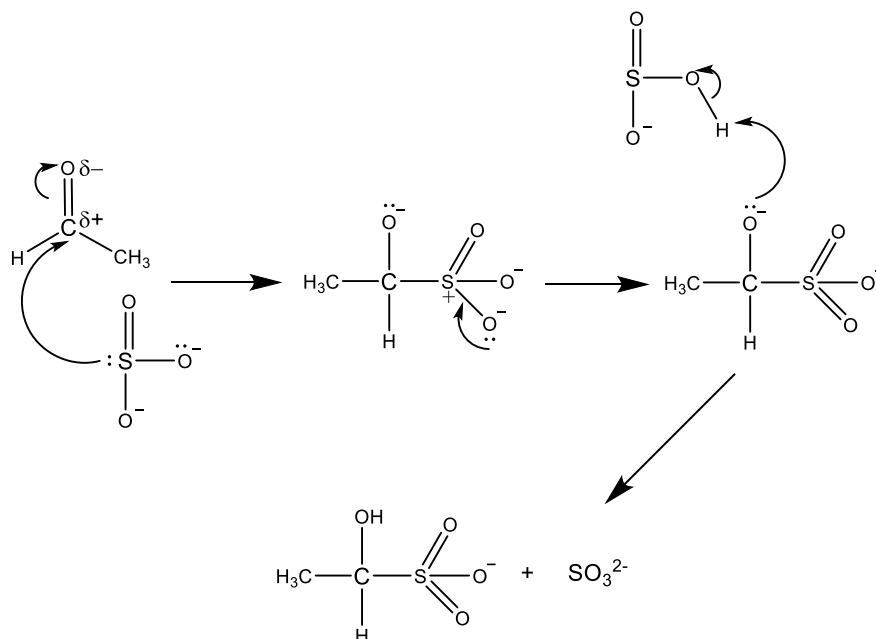
$$\text{Amount of } \text{SO}_3^{2-} \text{ produced} = \frac{25.0}{1000} \times 0.600 = 0.01500 \text{ mol}$$

$$\text{pH} = \text{pK}_a + \lg \frac{[\text{salt}]}{[\text{acid}]} = -\lg(6.73 \times 10^{-8}) + \lg \frac{[0.015/V]}{[0.01/V]} = 7.348 = 7.35$$

(Many candidates obtained the correct answer. The Henderson-Hasselbalch equation was well known but some weaker responses did not use the data given to calculate the equilibrium amounts of HSO_4^- and SO_3^{2-} . Some others did not appreciate that the initial amount of NaOH was equal to the equilibrium amount of SO_3^{2-} .)

[3]

(iv)



(This was generally well answered; most candidates applied their knowledge to suggest a good mechanism. Common errors were an omission of a curly arrow or an error in one of their structures.)

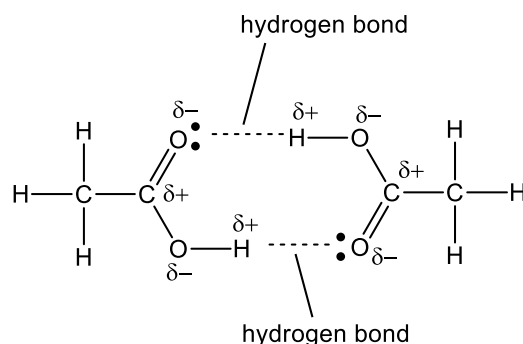
[3]

- (d) (i) In a reversible reaction, dynamic equilibrium is reached when the rate of forward reaction equals to the rate of backward reaction, where there is not net change in concentration of reactants and products/ where [reactant] and [product] remain constant.

(Many candidates gave an incomplete answer here. Most candidates stated that the rates of the forward and reverse reactions are equal. However, fewer gave the consequence of this.)

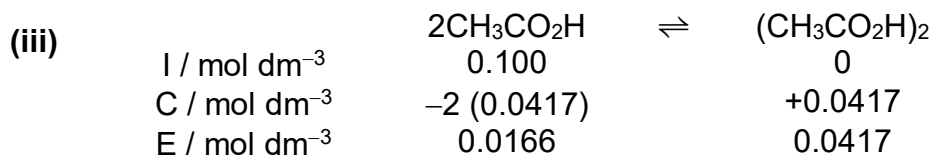
[1]

(ii)



(Candidates performed well on this question. Common errors were only drawing one hydrogen bond and omitting a partial charge on the oxygen atom involved in the hydrogen bond.)

[1]



$$\frac{[\text{CH}_3\text{CO}_2\text{H}]}{[(\text{CH}_3\text{CO}_2\text{H})_2]} = \frac{0.0417}{0.0166} = 2.51$$

Most candidates were able to calculate the correct ratio of $[(\text{CH}_3\text{CO}_2\text{H})_2] / [\text{CH}_3\text{CO}_2\text{H}]$.

[1]

(iv) $\Delta G^\ominus_{\text{rxn}}$ (aqueous solvent) will be less negative.

Water molecules forms stronger hydrogen bonds with ethanoic acid molecules as compared to weaker instantaneous dipole-induced dipole forces when non-polar solvent is used.

Ethanoic acid molecules will form favourable hydrogen bonds with large excess of water molecules. Hence, dimerisation is less favoured, leading to a less negative $\Delta G^\ominus_{\text{rxn}}$ (aqueous solvent).

OR

Energy level of aqueous solution of ethanoic acid is lower than non-polar solutions of ethanoic acid. The enthalpy change from aqueous solution of monomer to dimer will be smaller than enthalpy change from non-polar solution of monomer to dimer, leading to a less negative $\Delta G^\ominus_{\text{rxn}}$ (aqueous solvent).

OR

P.O.E lies to the left as the monomer ethanoic acid monomers forms hydrogen bonds with water and is energetically favoured. K_c will be lower leading to a less negative $\Delta G^\ominus_{\text{rxn}}$ (aqueous solvent).

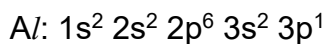
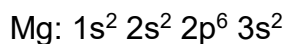
(This question was found difficult. Weaker responses stated that $\Delta G^\ominus$ would be lower or higher which is ambiguous.)

[1]

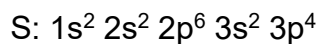
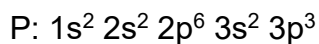
[Total: 20]

5 (a) The 1st ionisation energies **generally increase** across Period 3.

Across a period, **nuclear charge increases**, while there is **negligible increase in shielding effect** as electrons are removed from the same shell (the number of inner principal quantum shells of electrons remains the same). Hence, the **effective nuclear charge increases**. The **atomic radius** also **decreases**. Therefore, **valence electrons are increasingly more attracted by the positive nucleus** and more energy is required to remove the valence electrons.



1st IE of Al is lower than that of Mg. The electron removed from Al is a 3p electron whereas the electron removed from Mg is a 3s electron. Since the **3p electron is at a higher energy level** (further away from the nucleus) than the 3s electron, less energy is required to remove it from Al.



1st IE of S is lower than that of P. **Coulombic repulsion between the paired 3p electrons in S makes it easier to remove one of the paired 3p electrons** than an unpaired 3p electron from P. As a result, less energy is required.

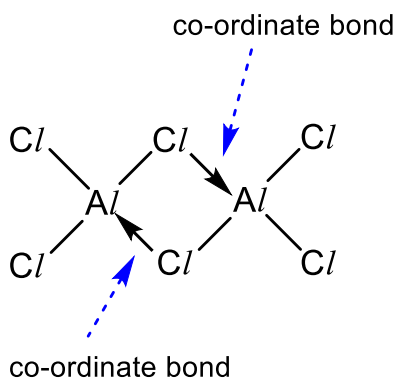
[3]

- (b) (i) $\text{AlCl}_3(\text{s})$ **dissolves readily in water** to form an acidic solution of **pH 3**. (Both hydration of ions and substantial hydrolysis of $\text{Al}^{3+}(\text{aq})$ occurs. Due to its high charge density, Al^{3+} is highly polarising and weakens the O–H bonds in the water molecules of the complex, causing the O–H bonds to break and hence release the hydrogen ions).

$\text{Al}_2\text{O}_3(\text{s})$ is **insoluble in water** and the solution has a **pH 7**. (The energy released during hydration of the ions (through the formation of ion-dipole interaction) is insufficient to compensate for the energy required to overcome the strong electrostatic forces of attraction between the Al^{3+} ions and O^{2-} ions.)

[2]

(ii)

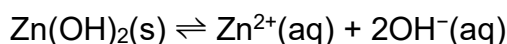


[1]

- (c) (i) $K_{\text{sp}} = [\text{Zn}^{2+}][\text{OH}^-]^2$ units: $\text{mol}^3 \text{dm}^{-9}$

[1]

- (ii) Let x be the solubility of $\text{Zn}(\text{OH})_2$



$$x(2x)^2 = 4x^3 = 2.0 \times 10^{-17}$$

$$x = 1.71 \times 10^{-6}$$

$$[\text{OH}^-] = 2x = \underline{\underline{3.42 \times 10^{-6}}} \text{ mol dm}^{-3}$$

[3]

$$\text{pOH} = -\lg 3.42 \times 10^{-6} = \underline{5.47}$$

$$\text{pH} = 14 - 5.47 = \underline{8.53}$$

- (iii) By adding $\text{HCl}(\text{aq})$, $[\text{OH}^-]$ **decreases** due to neutralisation with the acid and the **position of equilibrium in $\text{Zn}(\text{OH})_2(\text{s}) \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$ shifts to the right, increasing the solubility** of $\text{Zn}(\text{OH})_2$.

By adding $\text{ZnCl}_2(\text{aq})$, $[\text{Zn}^{2+}]$ **increases** (due to common ion effect) and the **position of equilibrium in $\text{Zn}(\text{OH})_2(\text{s}) \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$ shifts to the left, decreasing the solubility** of $\text{Zn}(\text{OH})_2$. [2]

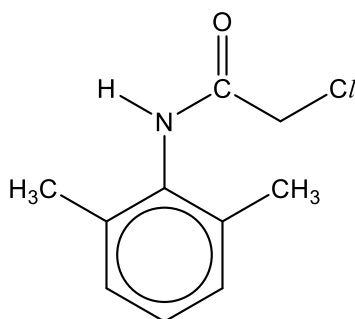
- (d) (i) Overall equation: $2\text{MnO}_2(\text{s}) + \text{Zn}(\text{s}) \rightarrow \text{Mn}_2\text{O}_3(\text{s}) + \text{ZnO}(\text{s})$

$$E^\ominus_{\text{cell}} = +0.15 - (-1.28) = \underline{+1.43 \text{ V}} \quad [1]$$

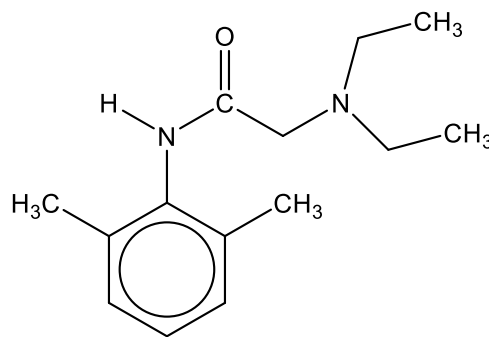
- (ii) $\Delta G^\ominus = -nFE^\ominus_{\text{cell}} = -(2)(96500)(1.43)$
 $= \underline{-276000 \text{ J mol}^{-1}}$ (3 s.f.)
 $= \underline{-276 \text{ kJ mol}^{-1}}$ (3 s.f.) [1]

- (e) (i) 2,6-dimethylphenylamine [1]

- (ii)



Compound T



Compound U

[2]

- (iii) Step 1: $\text{C}/\text{CH}_2\text{COC}/$

Step 2: excess ethanolic $\text{CH}_3\text{CH}_2\text{NHCH}_2\text{CH}_3$, heat in sealed tube [2]

- (iv) acid-base reaction / neutralisation [1]

[Total: 20]